# SSC MTS 2024 question paper with answers

**7 October 2024 · Shift 3**

75 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 15 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

## Numerical and Mathematical Ability 14 questions

**Q.1**

A man covers the first half of the journey at the speed of 60 km/h and the remaining at the speed of 84 km/h. What is his average speed (in km/h)?

**(A)** 75 **(B)** 70
**(C)** 65 **(D)** 60

**Q.2**

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 66 kg and 37 kg is:

**(A)** 49.6 **(B)** 48.6
**(C)** 50.6 **(D)** 51.6

**Q.3**

A person has to cover a distance of 8 km in 1 hour. If he covers one-fourth of the distance in one-third of the total time, then what should his speed (in km/h) be to cover the remaining distance in the remaining time so that the person reaches the destination exactly on time?

**(A)** 6 **(B)** 8
**(C)** 9 **(D)** 7

**Q.5**

A man bought a bullock and a cart for ₹10,000. He sold the bullock at a gain of 30% and the cart at a loss of 20%, thereby gaining 2% on the whole. Find the cost of the bullock.

**(A)** ₹4,600 **(B)** ₹4,400
**(C)** ₹5,600 **(D)** ₹5,400

**Q.6**

The mean of the first 8 odd natural numbers is:

**(A)** 5 **(B)** 6
**(C)** 8 **(D)** 7

**Q.8**

A closed cylindrical tank of height 280 cm and base radius 70 cm is to be made from a metal sheet. The required sheet for the same, in $m^2$, is:

**(A)** 13.8 **(B)** 14.4
**(C)** 12.8 **(D)** 15.4

**Q.9**

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 4% per year rate of interest for 3 years.

**(A)** 300
**(B)** 400
**(C)** 360
**(D)** 320

**Q.11**

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 3% per year rate of interest for 2 years.

**(A)** 150
**(B)** 250
**(C)** 170
**(D)** 210

**Q.12**

In view of Diwali, Rajnee, an electronics shop owner, offers a discount scheme to her customers that on buying 15 decorative bulbs, they get 5 decorative bulbs free. What is the effective percentage discount given by Rajnee to her customers?

**(A)** 33%
**(B)** 20%
**(C)** 30%
**(D)** 25%

**Q.13**

The price of 24 movie tickets is ₹768. How much (in ₹) will 32 movie tickets cost? [Assume that all tickets here are equally priced]

**(A)** 992
**(B)** 1040
**(C)** 1024
**(D)** 1056

**Q.15**

2 kg of copper at ₹15 per kg is mixed with 3 kg of zinc at ₹20 per kg. Find the average price of the mixture.

**(A)** ₹18
**(B)** ₹17.5
**(C)** ₹18.5
**(D)** ₹17

**Q.16**

A man spends 12% of his income on food, 18% on children's education, and 70% of the remaining on house rent. What percentage of his income is he left with?

**(A)** 28%
**(B)** 30%
**(C)** 21%
**(D)** 20%

**Q.17**

How many kilograms of tea, costing ₹250 per kg must be mixed with 25 kg of tea costing ₹400 per kg so that there may be a profit of 10% on selling the mixture at ₹330 per kg?

**(A)** 45 kg
**(B)** 50 kg
**(C)** 40 kg
**(D)** 60 kg

**Q.20**

Find the greatest number that exactly divides 12, 24 and 40.

**(A)** 2
**(B)** 40
**(C)** 4
**(D)** 3

## Reasoning Ability and Problem Solving 15 questions

**Q.2**

NGMI is related to LEKG in a certain way based on the English alphabetical order. In the same way, QJPL is related to OHNJ. To which of the following is TMSO related following the same logic?

**(A)** RKMQ
**(B)** KRMQ
**(C)** KEQM
**(D)** RKQM

**Q.3**

49 is related to 96 following a certain logic. Following the same logic, 54 is related to 106. To which ofthe following is 36 related following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 canbe performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3is not allowed.)

**(A)** 56
**(B)** 70
**(C)** 78
**(D)** 62

**Q.4**

Which two numbers should be interchanged to make the given equation correct? 52 + 44 – 38÷ 2 + 13× 4 – 26 = 97 (Note: Interchange should be done of entire number and not individual digits of a given number)

**(A)** 52 and 38
**(B)** 38 and 26
**(C)** 44 and 26
**(D)** 44 and 38

**Q.5**

What should come in place of the question mark (?) in the given series based on the English alphabetical order? ABC, EFG, IJK, MNO ,?

**(A)** STU
**(B)** UVW
**(C)** QRS
**(D)** TUV

**Q.6**

In a certain code language, ‘All men fight’ is coded as ‘AO EI UH’ and ‘men shave everyday’ is coded as ‘MN TP EI’. How is ‘men’ coded in that code language?

**(A)** UH
**(B)** EI
**(C)** TP
**(D)** AO

**Q.7**

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: A. No caps are hats. B. Some hats are helmets. Conclusions: I. All helmets are caps. II. Some helmets are hats.

**(A)** Only conclusion I follows.
**(B)** None of the conclusions follow.
**(C)** Both the conclusions follow.
**(D)** Only conclusion II follows.

**Q.8**

Select the number from among the given options that can replace the question mark (?) in the following series. [6/5], [3/2], [9/5],[21/10], [12/5], ?

**(A)** 5/2
**(B)** 27/10
**(C)** 9/2
**(D)** 29/10

**Q.9**

In a certain code language, ‘CURD’ is coded as ‘2436’ and ‘DOGS’ is coded as ‘3719’. What is the code for ‘D’ in the given code language?

**(A)** 3
**(B)** 1
**(C)** 6
**(D)** 2

**Q.10**

In a certain code language, ‘find the key’ is written as ‘so hd sm’ and ‘key to success’ is writtenas ‘gj so la’. How is ‘key’ writtenin the given language?

**(A)** la
**(B)** sm
**(C)** so
**(D)** gi

**Q.12**

HPML is related to MURQ in a certain way based on the English alphabetical order. In the same way, CJGF is related to HOLK. To which of the following is RZWV related, following the same logic?

**(A)** WEBA
**(B)** ABWE
**(C)** WEAB
**(D)** ABEW

**Q.13**

In a certain code language, ‘HOME’ is coded as ‘7519’ and ‘MUST’ is coded as ‘6782’. What is the code for ‘M’ in that language?

**(A)** 2
**(B)** 1
**(C)** 6
**(D)** 7

**Q.14**

A, B, C, D, E, F and G are sitting around a circular table facing the centre (but not necessarily in the same order). C sits second to the left of E, who is third to the left of A. G sits third to the right of B, who is not an immediate neighbour of F. D sits second to the right of B. How many people sit between D and A when counted from the left of D?

**(A)** Three
**(B)** Two
**(C)** One
**(D)** Four

**Q.15**

What will come in the place of the question mark (?) in the following equation, if ‘+’ and ‘–’ are interchanged and ‘×’ and ‘÷’ are interchanged? $2 \div 70 \times 2 - 30 + 30 = ?$

**(A)** 69
**(B)** 70
**(C)** 68
**(D)** 66

**Q.16**

In the following number-pairs, the second number is obtained by applying certain mathematical operations to the first number. Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 27, 109; 31, 125

**(A)** 35, 144
**(B)** 35, 145
**(C)** 36, 144
**(D)** 36, 145

**Q.18**

Eight friends, J, K, L, M, N, O, P and Q are sitting around a round table facing the centre (but not necessarily in the same order). K is an immediate neighbour of both Q and M. N is an immediate neighbour of both P and J. L is sitting immediately right to J. K is sitting at the immediate left of Q and is second to the right of O. Who is sitting to the immediate right of M?

**(A)** K
**(B)** O
**(C)** J
**(D)** P

## General Awareness 23 questions

**Q.1**

A person having stunted growth and discoloured hair is most probably suffering from _____.

**(A)** Glucose Deficiency
**(B)** Lipid Deficiency
**(C)** Vitamin A Deficiency
**(D)** Protein Deficiency

**Q.2**

Who among the following was the chief guest of India's first Republic Day in 1950?

**(A)** Rana Abdul Hamid
**(B)** Georgy Zhukov
**(C)** Queen Elizabeth
**(D)** Sukarno

**Q.3**

Which of the following dance forms is primarily associated with Eastern India states like Odisha, Jharkhand and West Bengal?

**(A)** Kuchipudi
**(B)** Chhau
**(C)** Kathak
**(D)** Bharatanatyam

**Q.4**

In the case of reptiles, their heart is not properly divided in to four chambers i.e., it consists of two auricles and an incompletely divided ventricle. Which of the following is an example of a reptile having a four-chambered heart?

**(A)** Turtles
**(B)** Snake
**(C)** Crocodile
**(D)** Lizard

**Q.5**

Maitraka dynasty was founded in which part of India?

**(A)** Eastern
**(B)** Western
**(C)** Southern
**(D)** Northern

**Q.6**

Which of the following is the autobiography of Bhishma Sahni?

**(A)** Aaj ke Ateet
**(B)** Aatma Parichaya
**(C)** Pankhhin
**(D)** Aarhdakatha

**Q.7**

Which of the following crops is grown in the Kharif season in India?

**(A)** cucumber
**(B)** Soyabean
**(C)** Barley
**(D)** Mustard

**Q.8**

Samrat Choudhary and Vijay Kumar Sinha became the Deputy Chief Minister of _____ on 28 January 2024.

**(A)** Bihar
**(B)** West Bengal
**(C)** Jharkhand
**(D)** Odisha

**Q.9**

Which of the following was the first Regional Rural Bank of India?

**(A)** Bandhan Bank
**(B)** Utkarsh Bank
**(C)** Vishwas Bank
**(D)** Pratham Bank

**Q.10**
In which of the following years was the 'Quit India Movement' launched?

**(A)** 1919
**(B)** 1935
**(C)** 1942
**(D)** 1920

**Q.11**
The elements that are liquid at room temperature (25°C) are:

**(A)** bromine and mercury
**(B)** bromine and chlorine
**(C)** helium and mercury
**(D)** helium and chlorine

**Q.12**
Sonu Nigam was awarded the Padma Shri in 2022 for which of the following fields?

**(A)** Art
**(B)** Sports
**(C)** Public Affairs
**(D)** Civil Service

**Q.13**
In which year was the Multi-dimensional Poverty Index launched in India?

**(A)** 2022
**(B)** 2020
**(C)** 2021
**(D)** 2023

**Q.14**
For transporting water from one place to another, some man-made structures used to distribute water to a place far away are known as:

**(A)** aquifuge
**(B)** aqueducts
**(C)** aquifer
**(D)** aquiclude

**Q.16**
Valabhi type of Nagara temples are generally _____ in shape, with a roof that rises into a vaulted chamber.

**(A)** circular
**(B)** triangular
**(C)** square
**(D)** rectangular

**Q.17**
Which team won the Ultimate Kho-Kho League Season 2 event?

**(A)** Mumbai Khiladis
**(B)** Chennai Quick Guns
**(C)** Gujarat Giants
**(D)** Telugu Yoddhas

**Q.19**
Who was the chairman of the drafting committee of the Constituent Assembly of India?

**(A)** Rajendra Prasad
**(B)** JB Kripalani
**(C)** Bhimrao Ambedkar
**(D)** Vallabhbhai Patel

**Q.20**
Who among the following was the first Indian representative in modern Olympics?

**(A)** Nilima Ghose
**(B)** Milkha Singh
**(C)** KD Jadhav
**(D)** Norman Pritchard

**Q.21**
Which of the following awards is NOT sponsored by the Madhya Pradesh government?

**(A)** Tagore Ratna
**(B)** Devi Ahilyabai Award
**(C)** Kishore Kumar Samman
**(D)** Tulshi Samman

**Q.22**

The Durg-Bastar-Chandrapur iron ore belt lies in which of the following groups of states?

**(A)** Madhya Pradesh - Gujarat
**(B)** Karnataka - Goa
**(C)** Chhattisgarh - Maharashtra
**(D)** Odisha - Jharkhand

**Q.23**

Which Articles of the Constitution of India provide for a parliamentary form of government at the Centre?

**(A)** Articles 74 and 75
**(B)** Articles 76 and 77
**(C)** Articles 71 and 72
**(D)** Articles 79 and 80

**Q.24**

Article _____ A supplements the 11th fundamental duty of the Indian Constitution?

**(A)** 28
**(B)** 33
**(C)** 21
**(D)** 46

**Q.25**

Who among the following founded the Saluva dynasty of Vijayanagara empire?

**(A)** Narsimha Deva Raya
**(B)** Bukka Raya I
**(C)** Rama Deva Raya
**(D)** Krishanadeva Raya

## English Language and Comprehension 23 questions

**Q.1**

Select the most appropriate option that can substitute the underlined word(s) in the given sentence. Manik said that it is mandatory that “each student pay” his/her tuition fees every month.

**(A)** each student paid
**(B)** each students pay
**(C)** each student paids
**(D)** each student pays

**Q.2**

Identify the INCORRECTLY spelt word in the given sentence. The microscopic chip that emits a uniqe signal has been embedded into your arm.

**(A)** uniqe
**(B)** emits
**(C)** embedded
**(D)** microscopic

**Q.3**

Select the INCORRECTLY spelt word.

**(A)** Always
**(B)** Until
**(C)** Altogether
**(D)** Fulfeel

**Q.4**

In the following sentence the underlined part contains an error. Alternatives to the underlined part are given as options. Select the correct alternative. I want to buy the “less” expensive bike, but with the best mileage.

**(A)** little
**(B)** lesser
**(C)** least
**(D)** most less

**Q.5**

Select the option that can be used as a one-word substitute for the given group of words. A person who rules with absolute power and authority

**(A)** Dictator
**(B)** Ruler
**(C)** Sovereign
**(D)** Senator

**Q.6**

Select the most appropriate idiom that can substitute the underlined segment in the given sentence. Bheem has been shown in the popular culture as someone who “eats in large quantity”.

**(A)** chews the fat
**(B)** eat humble pie
**(C)** eat like a bird
**(D)** eats like a horse

**Q.7**

Select the most appropriate synonym of the given word. Benevolent

**(A)** Enthusiastic
**(B)** Hostile
**(C)** Kind
**(D)** Mysterious

**Q.8**

The following sentence has been divided into four segments. Identify the segment that has a grammatical error. The thieves had no alternative / but to acquiesce with / the orders of / the police.

**(A)** The thieves had no alternative
**(B)** the police.
**(C)** but to acquiesce with
**(D)** the orders of

**Q.9**

Select the most appropriate option that can substitute the underlined segment in the given sentence. The most effective way to enhance your vocabulary is to “look to” unknown words in a dictionary.

**(A)** write off
**(B)** set by
**(C)** stand against
**(D)** look up

**Q.10**

Select the most appropriate option that can substitute the underlined segment in the given sentence. Once the seminar was over, the host “gave up” the certificates to the active participants.

**(A)** brought about
**(B)** put across
**(C)** gave away
**(D)** came by

**Q.11**

Rectify the error in the given sentence by selecting the correct spelling from the options provided. The nights of King Arthur were known for their chivalry.

**(A)** naughts
**(B)** knights
**(C)** knots
**(D)** nites

**Q.13**

Select the most appropriate option to fill in the blank. The teacher asked us to _____ the main idea of the passage in the exam.

**(A)** underestimate
**(B)** plagiarise
**(C)** summarise
**(D)** conclusion

**Q.14**

Select the most appropriate synonym of the given word. Careful

**(A)** Captivating
**(B)** Corrosive
**(C)** Cautious
**(D)** Corrupt

**Q.15**

Select the most appropriate ANTONYM of the underlined word in the given sentence. He is known for his “humility”.

**(A)** Pride
**(B)** Beauty
**(C)** Agility
**(D)** Knowledge

**Q.16**

Select the most appropriate option to fill in the blank. Poonam used a rubber band to keep the pencils together. 'Band' here means _____.

**(A)** a long rigid piece of wood or metal

**(B)** a range of values or frequencies within a series

**(C)** a thin ring of rubber used for holding things together

**(D)** a tube filled with water

**Q.18**

Select the appropriate option that can substitute the underlined idiom in the given sentence. The entire audience “was all ears” when the Indian-born American economist started giving a speech.

**(A)** was willing to exit the show

**(B)** was eager to speak plainly

**(C)** was arguing in favour of both sides

**(D)** was keen to listen

**Q.19**

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. Despite the challenges,/ they ultimate achieved / their goal of launching / their startup.

**(A)** their startup.

**(B)** they ultimate achieved

**(C)** Despite the challenges,

**(D)** their goal of launching

**Q.20**

Select the most appropriate option that can substitute the underlined segment in the given sentence. The head coach declared that the player “who will score the most points” in the game will clinch the trophy this year.

**(A)** who have score the maximum points

**(B)** who scores the maximum points

**(C)** who were score the maximum points

**(D)** who scores the points maximum

**Q.21**

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Three big fat rats once found a new use for _____1_____ tails. During the night, they came _____2_____ into the larder and found a jar with treacle in it. But the jar had a ____3___ neck, and try as they might, the rats could not reach the treacle either with their mouths or their paws. At last, a bright idea _____4_____ one of them. He climbed onto the jar and let his tail hang down into the treacle. Then he pulled it out nicely coated with treacle and allowed the two rats to lick it _____5_____.

Select the most appropriate option to fill in blank number 1.

**(A)** their

**(B)** the

**(C)** there

**(D)** it's

**Q.22**

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Three big fat rats once found a new use for _____1_____ tails. During the night, they came _____2_____ into the larder and found a jar with treacle in it. But the jar had a ____3___ neck, and try as they might, the rats could not reach the treacle either with their mouths or their paws. At last, a bright idea _____4_____ one of them. He climbed onto the jar and let his tail hang down into the treacle. Then he pulled it out nicely coated with treacle and allowed the two rats to lick it _____5_____.

Select the most appropriate option to fill in blank number 2.

**(A)** dreaming

**(B)** yelling

**(C)** creeping

**(D)** acting

**Q.23**

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Three big fat rats once found a new use for _____1_____ tails. During the night, they came _____2_____ into the larder and found a jar with treacle in it. But the jar had a ____3___ neck, and try as they might, the rats could not reach the treacle either with their mouths or their paws. At last, a bright idea _____4_____ one of them. He climbed onto the jar and let his tail hang down into the treacle. Then he pulled it out nicely coated with treacle and allowed the two rats to lick it _____5_____.

Select the most appropriate option to fill in blank number 3.

**(A)** slow
**(B)** narrow
**(C)** funny
**(D)** obese

**Q.24**

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Three big fat rats once found a new use for _____1_____ tails. During the night, they came _____2_____ into the larder and found a jar with treacle in it. But the jar had a ____3___ neck, and try as they might, the rats could not reach the treacle either with their mouths or their paws. At last, a bright idea _____4_____ one of them. He climbed onto the jar and let his tail hang down into the treacle. Then he pulled it out nicely coated with treacle and allowed the two rats to lick it _____5_____.

Select the most appropriate option to fill in blank number 4.

**(A)** stood
**(B)** struck
**(C)** flew
**(D)** went

**Q.25**

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Three big fat rats once found a new use for _____1_____ tails. During the night, they came _____2_____ into the larder and found a jar with treacle in it. But the jar had a ____3___ neck, and try as they might, the rats could not reach the treacle either with their mouths or their paws. At last, a bright idea _____4_____ one of them. He climbed onto the jar and let his tail hang down into the treacle. Then he pulled it out nicely coated with treacle and allowed the two rats to lick it _____5_____.

Select the most appropriate option to fill in blank number 5.

**(A)** at
**(B)** off
**(C)** in
**(D)** by

# Answer key

### Numerical and Mathematical Ability

| | | | | | |
|---|---|---|---|---|---|
| Q.1 **(B)** | Q.2 **(A)** | Q.3 **(C)** | Q.5 **(B)** | Q.6 **(C)** | Q.8 **(D)** |
| Q.9 **(C)** | Q.11 **(D)** | Q.12 **(D)** | Q.13 **(C)** | Q.15 **(A)** | Q.16 **(C)** |
| Q.17 **(B)** | Q.20 **(C)** | | | | |

### Reasoning Ability and Problem Solving

| | | | | | |
|---|---|---|---|---|---|
| Q.2 **(D)** | Q.3 **(B)** | Q.4 **(B)** | Q.5 **(C)** | Q.6 **(B)** | Q.7 **(D)** |
| Q.8 **(B)** | Q.9 **(A)** | Q.10 **(C)** | Q.12 **(A)** | Q.13 **(D)** | Q.14 **(D)** |
| Q.15 **(B)** | Q.16 **(D)** | Q.18 **(A)** | | | |

### General Awareness

| | | | | | |
|---|---|---|---|---|---|
| Q.1 **(D)** | Q.2 **(D)** | Q.3 **(B)** | Q.4 **(C)** | Q.5 **(B)** | Q.6 **(A)** |
| Q.7 **(B)** | Q.8 **(A)** | Q.9 **(D)** | Q.10 **(C)** | Q.11 **(A)** | Q.12 **(A)** |
| Q.13 **(C)** | Q.14 **(B)** | Q.16 **(D)** | Q.17 **(C)** | Q.19 **(C)** | Q.20 **(D)** |
| Q.21 **(A)** | Q.22 **(C)** | Q.23 **(A)** | Q.24 **(C)** | Q.25 **(A)** | |

### English Language and Comprehension

| | | | | | |
|---|---|---|---|---|---|
| Q.1 **(D)** | Q.2 **(A)** | Q.3 **(D)** | Q.4 **(C)** | Q.5 **(A)** | Q.6 **(D)** |
| Q.7 **(C)** | Q.8 **(C)** | Q.9 **(D)** | Q.10 **(C)** | Q.11 **(B)** | Q.13 **(C)** |
| Q.14 **(C)** | Q.15 **(A)** | Q.16 **(C)** | Q.18 **(D)** | Q.19 **(B)** | Q.20 **(B)** |
| Q.21 **(A)** | Q.22 **(C)** | Q.23 **(B)** | Q.24 **(B)** | Q.25 **(B)** | |

## Explanations

### Numerical and Mathematical Ability

**Q.1 · Answer (B) 70**

For two equal distances the average speed is $2 \times 60 \times 84/(60 + 84) = 10{,}080/144 = 70$ km/h.

**Q.2 · Answer (A) 49.6**

$(40 + 49 + 56 + 66 + 37)/5 = 248/5 = 49.6$ kg.

**Q.3 · Answer (C) 9**

One-fourth of 8 km is 2 km, covered in 20 minutes. The other 6 km must take the remaining 40 minutes (2/3 hour): $6 \div 2/3 = 9$ km/h.

**Q.5 · Answer (B) ₹4,400**

If the bullock cost x, $0.3x - 0.2(10{,}000 - x) = 2\%$ of ₹10,000 = 200, so $0.5x = 2{,}200$ and x = ₹4,400.

**Q.6 · Answer (C) 8**

The first n odd numbers add up to $n^2$, so the first 8 add up to 64 and their mean is $64/8 = 8$.

**Q.8 · Answer (D) 15.4**

A closed cylinder needs $2\pi r(h + r) = 2 \times 22/7 \times 70 \times (280 + 70) = 1{,}54{,}000$ cm² of sheet, which is 15.4 m².

**Q.9 · Answer (C) 360**

SI = $3{,}000 \times 4 \times 3/100 = ₹360$.

**Q.11 · Answer (D) 210**

SI = $3{,}500 \times 3 \times 2/100 = ₹210$.

**Q.12 · Answer (D) 25%**

The customer pays for 15 bulbs and gets 20, so 5 of every 20 are free: $5/20 = 25\%$.

**Q.13 · Answer (C) 1024**

One ticket costs $768 \div 24 = ₹32$, so 32 tickets cost $32 \times 32 = ₹1{,}024$.

**Q.15 · Answer (A) ₹18**

$(2 \times 15 + 3 \times 20)/(2 + 3) = 90/5 = ₹18$ per kg.

**Q.16 · Answer (C) 21%**

Food and education take 30%, leaving 70%. Rent takes 70% of that, 49% of the income, so $70 - 49 = 21\%$ is left.

**Q.17 · Answer (B) 50 kg**

To gain 10% at ₹330, the mixture must cost $330/1.1 = ₹300$ a kg. By alligation the teas at ₹250 and ₹400 go in the ratio $(400 - 300) : (300 - 250) = 2 : 1$, so $2 \times 25 = 50$ kg of the cheaper tea.

**Q.20 · Answer (C) 4**

The greatest number dividing all three is their HCF: $12 = 2^2 \times 3$, $24 = 2^3 \times 3$ and $40 = 2^3 \times 5$, so the HCF is $2^2 = 4$.

### Reasoning Ability and Problem Solving

**Q.2 · Answer (D) RKQM**

Each letter moves 2 places back: N → L, G → E, M → K, I → G, and QJPL → OHNJ likewise. So TMSO → RKQM.

**Q.3 · Answer (B) 70**

The second number is twice the first minus 2: $49 \times 2 - 2 = 96$ and $54 \times 2 - 2 = 106$. Likewise $36 \times 2 - 2 = 70$.

**Q.4 · Answer (B) 38 and 26**

Interchanging 38 and 26: $52 + 44 - 26 \div 2 + 13 \times 4 - 38 = 52 + 44 - 13 + 52 - 38 = 97$.

**Q.5 · Answer (C) QRS**
The clusters start at A, E, I and M, four letters apart, each followed by the next two letters. The next starts at Q: QRS.

**Q.6 · Answer (B) EI**
'men' is the only word in both sentences and 'EI' the only code in both, so men = EI.

**Q.7 · Answer (D) Only conclusion II follows.**
No cap is a hat, so the helmets that are hats cannot be caps: I does not follow. 'Some hats are helmets' gives 'some helmets are hats' (II).

**Q.8 · Answer (B) 27/10**
In tenths the terms are 12/10, 15/10, 18/10, 21/10 and 24/10, rising by 3/10 each time, so the next is 27/10.

**Q.9 · Answer (A) 3**
D is the only letter in both words and 3 the only digit in both codes, so D = 3.

**Q.10 · Answer (C) so**
'key' is the only word in both sentences and 'so' the only code in both, so key = so.

**Q.12 · Answer (A) WEBA**
Each letter moves 5 places forward, wrapping round after Z: H → M, P → U, M → R, L → Q, and CJGF → HOLK likewise. So R → W, Z → E, W → B and V → A: WEBA.

**Q.13 · Answer (D) 7**
M is the only letter in both words and 7 the only digit in both codes, so M = 7.

**Q.14 · Answer (D) Four**
Facing the centre, a person's left is clockwise. The only seating that fits is, clockwise from A: A, G, D, E, B, C, F (E third to A's left, C second to E's left, D second and G third to B's right, and B between E and C, away from F). Counting leftwards from D: E, B, C and F come before A, so four people sit between them.

**Q.15 · Answer (B) 70**
After interchanging the signs: 2 × 70 ÷ 2 + 30 − 30 = 70 + 30 − 30 = 70.

**Q.16 · Answer (D) 36, 145**
The second number is four times the first plus 1: 27 × 4 + 1 = 109 and 31 × 4 + 1 = 125. Likewise 36 × 4 + 1 = 145.

**Q.18 · Answer (A) K**
Facing the centre, a person's left is clockwise. Clockwise the order is K, M, O, L, J, N, P, Q: K is immediately left of Q and second to the right of O, with M as its other neighbour; L sits to J's right and N between J and P. The person to M's immediate right is K.

### General Awareness

**Q.1 · Answer (D) Protein Deficiency**
Stunted growth and discoloured hair are signs of kwashiorkor, caused by a lack of protein in the diet.

**Q.2 · Answer (D) Sukarno**
Sukarno, the first President of Indonesia, was the chief guest at India's first Republic Day on 26 January 1950.

**Q.3 · Answer (B) Chhau**
Chhau is performed in three styles named after their homes: Mayurbhanj in Odisha, Seraikela in Jharkhand and Purulia in West Bengal.

**Q.4 · Answer (C) Crocodile**
Crocodiles are the exception among reptiles: their heart has four fully separated chambers.

**Q.5 · Answer (B) Western**
The Maitraka dynasty was founded by Bhatarka in Saurashtra (Gujarat), with Valabhi as its capital: western India.

**Q.6 · Answer (A) Aaj ke Ateet**
'Aaj Ke Ateet' is the autobiography of the Hindi writer Bhisham Sahni, author of 'Tamas'; it was published after his death.

**Q.7 · Answer (B) Soyabean**
Soyabean is sown with the monsoon rains and harvested in autumn: a kharif crop. Barley and mustard are rabi crops, and cucumber is grown in the short zaid season.

**Q.8 · Answer (A) Bihar**
Samrat Choudhary and Vijay Kumar Sinha of the BJP became Bihar's Deputy Chief Ministers when Nitish Kumar was sworn in again as Chief Minister with BJP support on 28 January 2024.

**Q.9 · Answer (D) Pratham Bank**
Prathama Bank, set up at Moradabad in Uttar Pradesh on 2 October 1975 with Syndicate Bank as its sponsor, was India's first Regional Rural Bank.

**Q.10 · Answer (C) 1942**
The Quit India Movement was launched on 8 August 1942 at the Bombay session of the All India Congress Committee.

**Q.11 · Answer (A) bromine and mercury**
Bromine and mercury are the only elements that are liquid at 25°C; chlorine and helium are gases.

**Q.12 · Answer (A) Art**
The playback singer Sonu Nigam received the Padma Shri in 2022 in the field of Art.

**Q.13 · Answer (C) 2021**
NITI Aayog released India's National Multidimensional Poverty Index, a baseline report, in November 2021.

**Q.14 · Answer (B) aqueducts**
Aqueducts are man-made channels or bridges that carry water over long distances. An aquifer is a water-bearing layer of rock; an aquiclude holds water but does not let it flow, and an aquifuge neither holds nor passes it.

**Q.16 · Answer (D) rectangular**
Valabhi-type Nagara temples are rectangular buildings whose roof rises into a vaulted chamber.

**Q.17 · Answer (C) Gujarat Giants**
Gujarat Giants won Season 2 of Ultimate Kho Kho, beating Chennai Quick Guns 31-26 in the final at Cuttack in January 2024.

**Q.19 · Answer (C) Bhimrao Ambedkar**
Dr B. R. Ambedkar chaired the Drafting Committee of the Constituent Assembly; Rajendra Prasad was the Assembly's President.

**Q.20 · Answer (D) Norman Pritchard**
Norman Pritchard competed for India at the 1900 Paris Olympics and won two silver medals in athletics.

**Q.21 · Answer (A) Tagore Ratna**
The Tagore Ratna was a one-time honour of the Sangeet Natak Akademi, the national academy, marking Tagore's 150th birth anniversary. The Devi Ahilya, Kishore Kumar and Tulsi Samman are Madhya Pradesh government awards.

**Q.22 · Answer (C) Chhattisgarh - Maharashtra**
The Durg-Bastar-Chandrapur iron ore belt lies in Chhattisgarh and Maharashtra; the Bailadila hills of Bastar are part of it.

**Q.23 · Answer (A) Articles 74 and 75**
Articles 74 and 75 (a Council of Ministers to aid and advise the President, collectively responsible to the Lok Sabha) set up the parliamentary system at the Centre; Articles 163 and 164 do so in the states.

**Q.24 · Answer (C) 21**
The 86th Amendment (2002) added Article 21A, the right to free and compulsory education for children aged 6 to 14, and the 11th Fundamental Duty, Article 51A(k), for parents to provide them opportunities for education.

**Q.25 · Answer (A) Narsimha Deva Raya**
Saluva Narasimha Deva Raya, a general who took the throne from the last Sangama kings, founded the Saluva dynasty, the second of Vijayanagara's four dynasties.

### English Language and Comprehension

**Q.1 · Answer (D) each student pays**
SSC's key gives 'each student pays': 'each student' is singular and takes a singular verb. Formal English also allows the subjunctive 'pay' after 'it is mandatory that', as the sentence has it, but there is no 'no substitution' option; 'paid', 'students' and 'paids' are wrong.

**Q.2 · Answer (A) uniqe**
The correct spelling is 'unique'.

**Q.3 · Answer (D) Fulfeel**
The correct spelling is 'fulfil' ('fulfill' in American English).

**Q.4 · Answer (C) least**
The bike is chosen from all bikes, not between two, so the superlative is needed, matching 'the best mileage': 'the least expensive bike'.

**Q.5 · Answer (A) Dictator**
A dictator is a ruler with absolute power and authority.

**Q.6 · Answer (D) eats like a horse**
To 'eat like a horse' is to eat a great deal; to 'eat like a bird' is to eat very little.

**Q.7 · Answer (C) Kind**
'Benevolent' means well-meaning and kind.

**Q.8 · Answer (C) but to acquiesce with**
'Acquiesce' takes 'in' (or 'to'), not 'with': 'but to acquiesce in the orders of the police'.

**Q.9 · Answer (D) look up**
To 'look up' a word is to search for it in a dictionary.

**Q.10 · Answer (C) gave away**
To 'give away' prizes or certificates is to hand them out; to 'give up' is to stop or surrender.

**Q.11 · Answer (B) knights**
A knight is a mounted soldier of noble rank; 'nights' only sounds the same.

**Q.13 · Answer (C) summarise**
To 'summarise' is to state the main idea briefly; 'conclusion' is a noun and cannot follow 'to'.

**Q.14 · Answer (C) Cautious**
'Careful' means taking care to avoid harm: 'cautious'.

**Q.15 · Answer (A) Pride**
'Humility' is modesty; its opposite is 'pride'.

**Q.16 · Answer (C) a thin ring of rubber used for holding things together**
A rubber band is a thin ring of rubber used to hold things together.

**Q.18 · Answer (D) was keen to listen**
To be 'all ears' is to listen eagerly.

**Q.19 · Answer (B) they ultimate achieved**
'Achieved' is a verb, so it needs the adverb 'ultimately', not the adjective 'ultimate'.

**Q.20 · Answer (B) who scores the maximum points**
In a clause about the future that depends on another future ('will clinch'), English uses the simple present: 'the player who scores the maximum points'.

**Q.21 · Answer (A) their**
The tails belong to the rats, so the possessive 'their' is needed: 'a new use for their tails'. 'There' is an adverb and 'it's' means 'it is'.

**Q.22 · Answer (C) creeping**
The rats came 'creeping' into the larder at night, moving slowly and quietly.

**Q.23 · Answer (B) narrow**
The rats could not reach the treacle because the jar had a 'narrow' neck.

**Q.24 · Answer (B) struck**
An idea 'strikes' someone when it comes suddenly into their mind: 'a bright idea struck one of them'.

**Q.25 · Answer (B) off**
The two rats licked the treacle 'off' the tail.

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